Google Skills

The problem: The evolution that led us here

Introduction

Think about the last time you asked Gemini or any other AI to do something for you. Maybe you asked it to help plan a trip, manage your calendar, or analyze some data. It probably gave you great advice, like detailed steps, helpful suggestions, or even some sample code. But then you still had to do all the actual work yourself.

This gap between knowing and doing represents the next significant step in the evolution of AI. While we have incredibly intelligent systems capable of advising us on a vast array of tasks, the final execution still rests on our shoulders. It’s akin to having a world-class advisor who can’t pick up the phone, send an email, or press a button on your behalf.

Let's trace how we got here and each step’s limitations.

First, we had LLM APIs. These were revolutionary. Suddenly we could have conversations with AI, ask questions, and generate content. But try asking an LLM to check your email or book a flight. It can’t. The LLM is frozen in time and disconnected from the world. It can process information but can’t interact with external systems.

Then came LLMs with function calling. This seemed like the answer! Now the model could call functions, interact with APIs, and even update databases. But here's the catch: you still had to orchestrate everything. Every function call needed your explicit instruction. Want to book a complete trip? You'd need to manually coordinate dozens of function calls: search flights, check availability, compare prices, book flights, find hotels, book hotels, and arrange transportation. The model became a better assistant, but you were still the one doing all the thinking and planning.

Here’s what this looks like in practice:
Imagine asking “book me a trip to Paris next week.”

With an LLM, you get advice: “Here’s how to book a trip to Paris...” With function calling, you get capability but not autonomy: “I can search for flights if you tell me the exact dates, then I can search for hotels if you tell me the area, then I can...” and so on.

What you really want is: “I’ve found three flight options that work with your calendar, matched them with nearby hotels in your budget, and I’m ready to book the combination that works best. Should I proceed?”

That’s the promise of agents.

```
graph LR
   subgraph <b>Stage 1</b>"
       direction TB
       A1(Capability<br/><b>Text Only</b>)
       A2[Technology<br/><b>LLM API</b>]
   end

   subgraph "<b>Stage 2</b>"
       direction TB
       B1(Capability<br/><b>Can Use Tools</b>)
       B2[Technology<br/><b>LLM + Function Calling</b>]
   end


   subgraph "<b>Stage 3</b>"
       direction TB
       C1(Capability<br/><b>Pursues Goals</b>)
       C2[Technology<br/><b>AI Agent</b>]
   end

   A2 -->|<b>+ Functions</b>| B2
   B2 -->|<b>+ Autonomy</b>| C2

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   style B2 fill:#bbf,stroke:#333,stroke-width:2px
   style C2 fill:#bfb,stroke:#333,stroke-width:2px

   style A1 fill:#fff,stroke:#333,stroke-width:1.5px,stroke-dasharray: 5 5
   style B1 fill:#fff,stroke:#333,stroke-width:1.5px,stroke-dasharray: 5 5
   style C1 fill:#fff,stroke:#333,stroke-width:1.5px,stroke-dasharray: 5 5
```

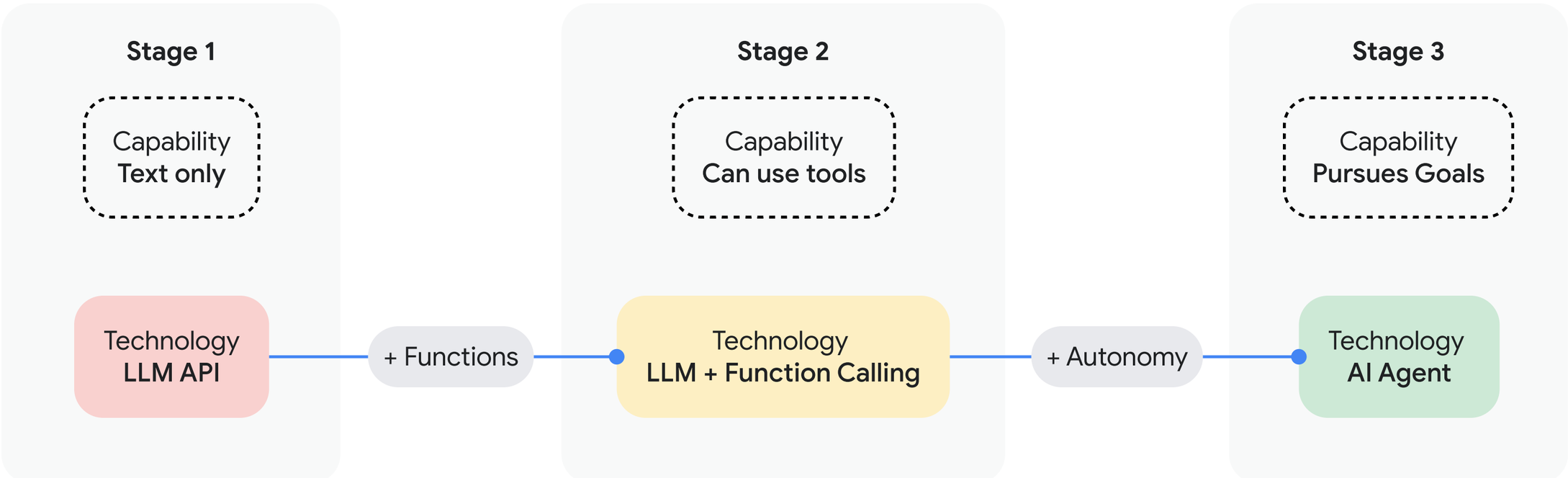

Figure: The evolution from LLMs to Agents — each step adds critical capabilities.